

Omninet AI System Architecture

Document – Version 1.0

Ver.	Date	Author	Description	Approved by	Approval Date
1.0	Mar 26, 2025	Architecture Team	Initial release for Omninete AI project		

1. Introduction

1.1 Scope

This document describes the system architecture for the Omninete AI Internet kiosk platform. It covers the edge (kiosk) deployment, the cloud backend, and the AI Support Center. The architecture is designed to meet the functional and non-functional requirements defined in the **Omninete AI Marketing Requirements Document (v1.2)** and the **Omninete AI System Requirements Document (v1.0)**.

1.2 Audience

- System architects and developers
- Testers and quality assurance teams
- Deployment and operations engineers
- Product managers and stakeholders

1.3 References

- [MRD] Omninet AI Marketing Requirements Document v1.2
- [SysRD] Omninet AI System Requirements Document v1.0
- [ContentPolicy] Omninet Content Filtering Policy (TBD)
- [AIGov] AI Governance Document (TBD)

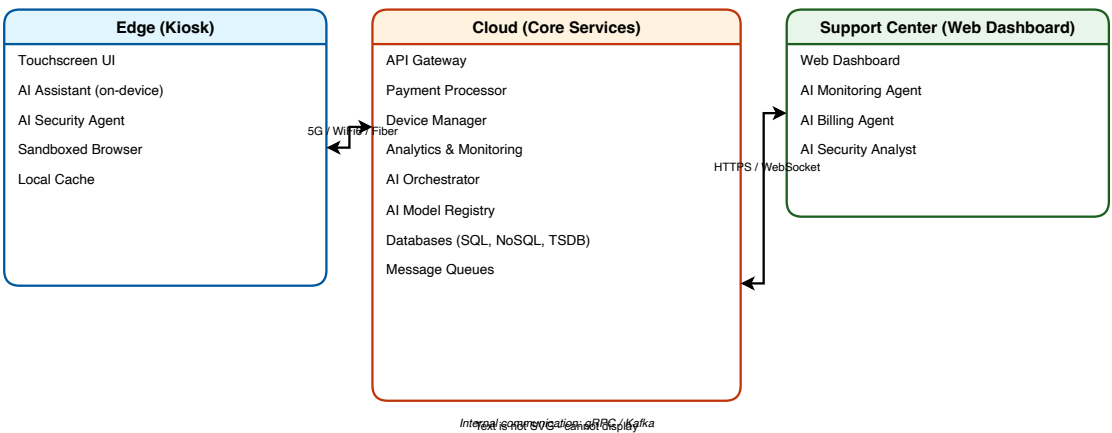
2. System Overview

Omninet AI is a distributed system consisting of three major tiers:

Tier	Description
Edge (Kiosk)	Physical kiosks deployed in public locations. They provide a touch UI, sandboxed browser, local AI agents, and secure payment processing.
Cloud (Core Services)	A cloud-native backend that manages device registration, payment processing, AI model orchestration, analytics, and central administration.
Support Center	A web-based dashboard for human agents, augmented by AI agents (monitoring, billing, security). It provides real-time visibility and control over the kiosk network.

! [High-level architecture diagram – see draw.io representation in Section 2.1]

2.1 High-Level Architecture Diagram (Conceptual)



All communication between tiers is encrypted (TLS 1.3). The edge connects to the cloud via a secure API gateway; the support center connects to the cloud via a web interface with real-time updates.

3. Edge (Kiosk) Deployment Design

3.1 Hardware Specifications (Baseline)

- **Processor:** Industrial-grade x86-64 or ARM-based (e.g., Intel NUC, Raspberry Pi 4+)
- **Memory:** 8 GB RAM minimum
- **Storage:** 128 GB SSD (for OS, cache, logs)
- **Display:** 24" industrial touchscreen (capacitive, IP65 rated)
- **Connectivity:** Integrated 5G modem, WiFi6, and Ethernet (Fiber backup)
- **Peripherals:** Bill acceptor, coin acceptor, NFC reader, magnetic stripe reader, thermal printer (optional)

3.2 Software Stack

Component	Technology / Description
Operating System	Ubuntu Linux 22.04 LTS or Windows 11 IoT Enterprise (container-host)
Container Runtime	Docker / Podman (for micro-services on edge)
Sandboxed Browser	Embedded Chromium (kiosk mode) with limited file system access, enforced by AppArmor / SELinux
AI Assistant	On-device lightweight LLM (e.g., TinyLlama) for common queries; falls back to cloud AI Orchestrator for complex tasks.
AI Security Agent	Python-based threat detection using on-device ML models (updated via cloud).

Component	Technology / Description
Local Cache	SQLite for session data, Redis for ephemeral state, file system for web cache.
Payment Module	Native integration with cash acceptor and NFC/EMV reader; communicates with cloud Payment Processor.
Update Client	Polls cloud for updates (OS patches, container images, AI models).

3.3 Security at the Edge

- **Hardware Security:** Trusted Platform Module (TPM) for secure boot and key storage.
- **Sandboxing:** Each user session runs in a separate container; browser runs with minimal privileges.
- **Data Wipe:** On session termination, the local cache and all user data are